## Data Quality Assessment Summary

A data quality assessment was conducted across all five event tables to identify missing values, duplicate records, logical inconsistencies, data type issues, and user IDs that could not be matched with the registration table.

| Table | Null values | Duplicate Rows | Logical Issue | Unmatched User IDs |
|---|---|---|---|---|
| sod_adrevenue | 1807 | 17 | None identified | 35,232 |
| sod_battle_short | 3458 | 56 | None identified | 48,891 |
| sod_register | 188 | 0 | None identified | N/A |
| sod_revenue | 93 | 0 | 22 negative revenue values | 2,007 |
| sod_session_start | 2199 | 0 | None identified | 80,621 |

# Key Findings and Actions Taken

**1. Missing Values**

- Null values were identified across all five tables.
- These records should be investigated to determine whether the missing fields affect analysis before deciding how to handle them.

**2. Duplicate Records**

- Duplicate records were identified in sod_adrevenue (17 records) and sod_battle_short (56 records).
- These records require validation before removal to avoid accidentally deleting legitimate events.

**3. Logical Inconsistencies**

- In sod_revenue, 22 records contain negative values in event_revenue_gross_usd.
- These records should be investigated to determine whether they represent refunds, reversals, or invalid values.

**4. Data Type Standardization**

- The user_install_time and event_time columns were converted to appropriate datetime data types across all five tables to support accurate time-based analysis**.**

**5. Referential Integrity**

- Several user IDs in the event tables could not be matched with sod_register.
- The highest number of unmatched IDs was found in sod_session_start (80,621), followed by sod_battle_short (48,891).
- These users should not automatically be treated as invalid, as they may belong to players whose registration records are missing from the available dataset.

# Key Metrics for Evaluating a Gaming App(Question: 1 and 3)

## 1. Session Duration

Session duration is an important engagement metric for a gaming app because it measures how much time players spend playing the game. It helps us understand player engagement and whether the game is successful in keeping players interested.

However, session duration alone cannot determine a game's success. We also need to evaluate other aspects, such as retention, gameplay activity, and monetization.

To understand session duration more effectively, I would analyze the following metrics:

- **Average Session Duration**: Measures the average time players spend in a session or across their sessions, depending on the definition used. However, the mean can be affected by outliers, such as a small number of players spending unusually long periods in the game.
- **Median Session Duration**: Represents the middle value when session durations are sorted. It is less affected by extreme values and helps us understand a typical player's experience.
- **Percentile Analysis**: Examining the 25th, 50th, 75th, 90th, and 99th percentiles helps us understand the distribution of session duration. It allows us to compare casual players with highly engaged players and identify unusually long sessions.

Together, these metrics provide a more complete picture of player engagement than the average alone.

## 2. Revenue and Player Monetization

Another important area is monetization, which helps us understand how players generate revenue for the game. By analyzing in-app purchase revenue and advertising revenue separately, we can identify different types of players and understand their contribution to the business. For example, we can segment players into:

- Non-monetizing players: Players who have not generated any recorded revenue.
- Ad-monetizing players: Players who generate revenue through rewarded advertisements but have not made in-app purchases.
- Paying players: Players who generate revenue through in-app purchases.
- High-value players: Players who contribute a relatively large amount of revenue through purchases, advertising, or both.

This segmentation helps us understand which player groups contribute most to revenue and how their engagement differs.

We can also analyze metrics such as total revenue, average revenue per paying user (ARPPU), revenue per installed user (RPI), and the percentage of players who make purchases.

Importantly, paying players are not automatically more valuable than ad-monetizing players. Their relative value depends on the revenue they generate, the costs involved, and how consistently they remain engaged.

| | |
|---|---|
| Session Duration per user AVG | 114 |
| Session Duration per user MEDIAN | 26 |
| Session Duration MAX | 19,478 |
| Session Duration per user PERCENTILE (90) | 293 |
| Session Duration per user PERCENTILE (99) | 1,323 |
| Revenue AVG | 0 |
| Revenue MEDIAN | 0 |
| Revenue MAX | 860 |
| Revenue PERCENTILE (98.2) | 1 |
| Revenue PERCENTILE (99) | 5 |
| Ad Revenue PERCENTILE (97) | 1 |
| Ad Revenue MAX | 21 |
| Ad Revenue AVG | 0 |
| Ad Revenue MEDIAN | 0 |
| Ad Revenue PERCENTILE 99 | 1 |

# Country-Level Revenue and Player Analysis (Question 2)

I analyzed the game's performance across different countries to understand how the player base and revenue are distributed geographically.

## Key Findings

### 1. The US is the leading revenue-generating market

The US is the only country contributing more than **40% of total revenue**, making it the most important market in terms of monetization. South Korea (KR) is the second-largest revenue contributor, accounting for **6.56% of total revenue**.

### 2. Russia has a large player base but lower revenue contribution

Russia (RU) is the second-largest market by user base, with **14,590 users**. However, it does not rank among the top revenue contributors (2.8%). This suggests a potential gap between user acquisition and monetization that warrants further investigation.

### 3. Revenue is highly concentrated among a small percentage of users

My percentile analysis indicates that the first positive revenue value appears around the **94th percentile**. This suggests that revenue may be concentrated among a small proportion of users, with approximately 6% of users potentially contributing to most of the revenue.

This is an important finding because it highlights the need to understand the behavior of high-value players and identify opportunities to improve monetization among the broader player base.

### 4. Session duration in the US

For US users, the average session duration is **136 minutes**, while the median is also **29 minutes**. Among paying users, the average session duration is **129 minutes**, but the median is **275 minutes**.

The difference between the average and median for paying users suggests that their session durations may be unevenly distributed, potentially with some shorter sessions pulling down the average.

| User Country | Total Revenue | Avg Total Rvenue | Total Ad Revenue | Total Iap Revenue | Unique user Count | Revenue Percentile Rank (Min Paying user) | Avg Session Duration | Median Session | Median session of paying user | Avg Session of Paying user | Revenue Contrubution |
|---|---|---|---|---|---|---|---|---|---|---|---|
| US | 27,868 | 1.10 | 4,744 | 23,124 | 25,347 | 0.946 | 136 | 29 | 129 | 275 | 40.53% |
| KR | 4,513 | 1.06 | 917 | 3,596 | 4,245 | 0.941 | 117 | 23 | 76 | 220 | 6.56% |
| CN | 4,157 | 0.46 | 1,764 | 2,392 | 9,065 | 0.863 | 91 | 18 | 137 | 269 | 6.05% |
| JP | 4,008 | 1.43 | 953 | 3,056 | 2,803 | 0.966 | 181 | 28 | 155 | 375 | 5.83% |
| DE | 3,349 | 0.57 | 925 | 2,423 | 5,894 | 0.893 | 118 | 24 | 101 | 237 | 4.87% |
| TW | 2,529 | 1.10 | 404 | 2,125 | 2,297 | 0.951 | 142 | 28 | 115 | 271 | 3.68% |
| GB | 2,121 | 0.51 | 325 | 1,796 | 4,126 | 0.873 | 113 | 26 | 118 | 246 | 3.08% |
| RU | 1,983 | 0.14 | 766 | 1,217 | 14,590 | 0.746 | 135 | 31 | 121 | 260 | 2.88% |
| FR | 1,828 | 0.55 | 234 | 1,594 | 3,306 | 0.883 | 116 | 26 | 116 | 230 | 2.66% |
| CA | 1,784 | 0.79 | 147 | 1,637 | 2,256 | 0.922 | 127 | 24 | 128 | 288 | 2.60% |
| AU | 1,415 | 0.93 | 108 | 1,307 | 1,523 | 0.932 | 120 | 26 | 129 | 264 | 2.06% |
| CH | 911 | 2.20 | 70 | 840 | 414 | 0.976 | 151 | 28 | 150 | 298 | 1.32% |
| NL | 845 | 0.63 | 58 | 786 | 1,349 | 0.898 | 110 | 25 | 93 | 224 | 1.23% |
| BR | 819 | 0.08 | 106 | 713 | 9,648 | 0.698 | 96 | 24 | 113 | 229 | 1.19% |
| SA | 622 | 0.42 | 28 | 594 | 1,486 | 0.854 | 77 | 22 | 102 | 207 | 0.90% |
| TH | 579 | 0.12 | 141 | 438 | 4,772 | 0.732 | 105 | 26 | 110 | 228 | 0.84% |
| KW | 563 | 1.95 | 8 | 555 | 288 | 0.971 | 102 | 25 | 140 | 301 | 0.82% |
| SE | 539 | 0.77 | 32 | 507 | 700 | 0.917 | 102 | 21 | 115 | 250 | 0.78% |
| TR | 520 | 0.11 | 65 | 455 | 4,708 | 0.727 | 116 | 29 | 121 | 258 | 0.76% |
| MY | 455 | 0.21 | 64 | 391 | 2,120 | 0.805 | 117 | 31 | 112 | 229 | 0.66% |
| IT | 452 | 0.17 | 96 | 356 | 2,645 | 0.795 | 126 | 30 | 109 | 247 | 0.66% |
| MX | 445 | 0.05 | 83 | 362 | 8,374 | 0.654 | 100 | 26 | 116 | 220 | 0.65% |
| HK | 430 | 0.73 | 43 | 387 | 593 | 0.907 | 108 | 23 | 103 | 235 | 0.63% |
| DJ | 390 | 43.33 | 0 | 390 | 9 | 1.000 | 171 | 31 | 201 | 397 | 0.57% |
| AT | 348 | 0.67 | 26 | 322 | 522 | 0.902 | 129 | 31 | 114 | 224 | 0.51% |
| PL | 329 | 0.14 | 113 | 216 | 2,339 | 0.751 | 122 | 27 | 99 | 251 | 0.48% |
| SG | 320 | 0.50 | 28 | 293 | 641 | 0.868 | 119 | 27 | 113 | 234 | 0.47% |
| IQ | 266 | 0.13 | 8 | 258 | 2,101 | 0.737 | 77 | 26 | 99 | 180 | 0.39% |
| KZ | 243 | 0.13 | 16 | 227 | 1,866 | 0.741 | 148 | 36 | 141 | 293 | 0.35% |
| ES | 230 | 0.11 | 69 | 161 | 2,116 | 0.722 | 124 | 28 | 109 | 259 | 0.33% |
| HU | 200 | 0.17 | 20 | 180 | 1,173 | 0.790 | 110 | 26 | 104 | 227 | 0.29% |
| UA | 195 | 0.04 | 90 | 105 | 4,398 | 0.634 | 122 | 28 | 111 | 246 | 0.28% |
| BE | 185 | 0.24 | 22 | 164 | 772 | 0.810 | 102 | 24 | 109 | 234 | 0.27% |
| NO | 184 | 0.54 | 23 | 161 | 340 | 0.878 | 124 | 25 | 142 | 254 | 0.27% |
| PH | 183 | 0.05 | 43 | 141 | 3,763 | 0.639 | 106 | 29 | 111 | 227 | 0.27% |
| CZ | 175 | 0.16 | 25 | 150 | 1,097 | 0.771 | 97 | 22 | 95 | 215 | 0.25% |
| VN | 174 | 0.04 | 42 | 133 | 4,475 | 0.615 | 76 | 21 | 88 | 196 | 0.25% |
| ID | 174 | 0.04 | 53 | 122 | 4,170 | 0.624 | 125 | 32 | 125 | 248 | 0.25% |

# Marketing vs. Organic User Segmentation(Question: 2)

I analyzed player engagement by acquisition source to compare users acquired through marketing campaigns with those who joined organically.

## Key Findings

### 1. Higher Session Duration

The average session duration of marketing-acquired users is approximately **60% higher than that of organic users**. This suggests that users acquired through marketing campaigns spend more time in the game.

### 2. Higher Median Session Duration

The median session duration for marketing-acquired users is **47 minutes, approximately 2× the median for organic users**. Since the median is less affected by extreme values, this indicates that the typical marketing-acquired user also spends more time playing.

**3. Marketing Contribution Relative to User Base**

Considering the size of the marketing-acquired user base alongside its engagement levels, marketing appears to be attracting an engaged audience. This is a promising signal for the effectiveness of the acquisition strategy.

Marketing wise Segemnt

| | User Marketing User | |
|---|---|---|
| | False | True |
| Total Revenue | 44,381 | 24,371 |
| Total Ad Revenue | 9,926 | 3,115 |
| Total Iap Revenue | 34,455 | 21,257 |
| AVG session per user | 104 | 160 |
| Total Revenue Contributi.. | 64.55% | 35.45% |
| Avg Seesion Paying users | 242 | 273 |
| Median session of paying .. | 109 | 146 |
| Median session | 24 | 47 |
| Avg Revenue per user | 0.29 | 0.78 |
| Unique user Count | 150,553 | 31,341 |

# Device-Level Segmentation: Android vs. iOS (Question: 2)

I analyzed revenue contribution, player distribution, and session duration across Android and iOS to understand which platform attracts more valuable players.

## Key Findings

**1. Revenue Contribution**

Android generates approximately **2× the total revenue** and **3× the advertising revenue** of iOS. Its in-app purchase revenue is approximately **1.5× higher** than iOS.

However, Android also has approximately **5× as many users** as iOS. Therefore, higher total revenue does not necessarily mean Android users are more valuable individually.

**2. Player Value**

Despite Android generating more total revenue, iOS users appear to deliver stronger monetization per user. This suggests that iOS may attract a more valuable user base on an individual basis.

To validate this conclusion, I would compare revenue per user (RPI), in-app purchase conversion, and average revenue per paying user (ARPPU) across both platforms.

**3. Session Duration Among Paying Users**

Paying users on iOS have approximately **1.5× the average session duration** of paying users on Android. Additionally, the average session duration of Android paying users is approximately **1.24× that of Android users overall**, based on my analysis.

These findings suggest that paying users tend to spend more time in the game, with iOS paying users showing particularly strong session engagement.

Device wise segment

| | User Device Platform | |
|---|---|---|
| | Android | iOS |
| Total Revenue | 42,793 | 25,959 |
| Total Ad Revenue | 9,262 | 3,779 |
| Total Iap Revenue | 33,531 | 22,180 |
| AVG session per user | 108 | 136 |
| Total Revenue Contributi.. | 62.24% | 37.76% |
| Avg Seesion Paying users | 238 | 295 |
| Median session of paying .. | 109 | 157 |
| Median session | 25 | 32 |
| Avg Revenue per user | 0.29 | 0.78 |
| Unique user Count | 148,522 | 33,372 |

# Segmentation: Device vs. Marketing (Question: 2)

As we can see, the average total revenue per user on iOS is relatively similar between marketing and organic users. However, on Android, marketing users generate an average total revenue of 74, which is considerably higher than that of organic users. This suggests that marketing has a stronger impact on revenue generation among Android users.

Although the average ad revenue follows a similar pattern across both acquisition channels, Android marketing users have an average session duration approximately 60% higher than organic users. The median session duration also shows a noticeable difference, indicating that the higher engagement is not driven solely by a few users with exceptionally long sessions.

Overall, these findings suggest that marketing-acquired Android users demonstrate stronger engagement and potentially better monetization than organic users. However, further analysis of retention, payer conversion, and acquisition costs would help confirm whether marketing delivers better overall user quality and return on investment.

## Marketing vs device

| | User Device Platform / User Marketing User | | | |
|---|---|---|---|---|
| | Android | | iOS | |
| | False | True | False | True |
| Total Revenue | 28,179 | 14,613 | 16,201 | 9,758 |
| Avg Total Rvenue | 0.22 | 0.74 | 0.74 | 0.84 |
| Total Ad Revenue | 7,105 | 2,156 | 2,820 | 958 |
| Total Iap Revenue | 21,074 | 12,457 | 13,381 | 8,800 |
| Unique user Count | 128,770 | 19,752 | 21,783 | 11,589 |
| Avg Session Duration | 100 | 160 | 124 | 159 |
| Median Session | 23 | 46 | 27 | 48 |
| Median session of paying user | 103 | 137 | 149 | 167 |
| Avg Session of Paying user | 232 | 265 | 299 | 289 |
| SUM([Total Revenue]) / TOTAL(SUM([Total Revenue])) | 40.99% | 21.26% | 23.56% | 14.19% |